ORION OFFICE DIRECT

E-Commerce Analytics
Project Brief

Dataset: e_commerce (Xóm Dataset · SQL Server) **Prepared by:** Business Analyst — Growth & Strategy **Date:** October 2026

STEP 1 — DATA UNDERSTANDING

ecom_sales	51,300	11	One order line item — a single SKU within one order	order_id, customer_id, order_date, sales, profit, discount, segment, region_code, product_code
customer	17,400	11	One customer demographic profile	customer_id, annual_income, education_level, occupation, gender, birth_date
product	28,400*	4	One product record — category/subcategory mapping	product_code, product, category, subcategory
region	3,800	8	One shipping destination at city level	region_code, city, country, region, market, lat/lon

* product table has 28 400 rows but only ~1 900 distinct product_codes — must verify uniqueness before any JOIN to avoid fan-out inflation.

Critical Data Quality Checks (run these before any analysis)

■ CRITICAL	**product_code uniqueness**	COUNT(*) vs COUNT(DISTINCT product_code) in product table. If not 1:1, every JOIN inflates SUM(sales).
■ CRITICAL	**segment is order-level, not customer-level**	A single customer_id may appear under multiple segments across orders. Never GROUP BY segment without first assigning a canonical segment per customer.
■ CRITICAL	**discount format**	Verify range: if 0.0–1.0 it is a rate; if 0–100 it is a percentage. Context implies rate. Run MIN/MAX/AVG to confirm.
■ WARN	**profit can be negative**	This is a business signal, not a data error. Never filter out negative-profit rows.
■ WARN	**annual_income currency**	No unit declared — assume USD. Must state this assumption in every deliverable.
■ WARN	**No ship_date / return flag**	Cannot compute logistics cycle time or net revenue after returns. Declare as a limitation.

STEP 2 — BUSINESS CONTEXT

Company	Orion Office Direct — Series C B2C/B2B e-commerce marketplace operating in 140+ countries across US, EMEA, APAC, LATAM. No brick-and-mortar presence; 100% web + app. Annual revenue ~$2.3M, gross margin ~20% (compressed by logistics costs and aggressive discounting). Catalog: 3 categories (Office Supplies, Furniture, Technology), ~1,900 SKUs. Customer segments: Consumer 60%, Corporate 25%, Home Office 15%.
My Role	Business Analyst, Growth & Strategy team. Sit across Growth, Finance, and Merchandising. Report directly to three stakeholders with competing priorities and a shared deadline: board review in 6 weeks.
Project Trigger	Marketing submitted a $400K discount campaign budget request for Q1 2026. The CFO blocked it with one line: 'Show me that historical discounting has produced net-positive margin — then I'll sign.' Simultaneously, the Head of Growth needs the top 3 markets for expansion investment before the same board review. Both decisions draw from the same dataset.

Stakeholders

Linda Park	CFO	Margin integrity. Suspects discount campaigns are destroying profit, not building it.	Is every dollar of discount producing more than a dollar of contribution margin?
Marcus Tan	Head of Growth	Repeat purchase rate and market expansion. Most customers buy once then disappear.	Which 3 markets do we double down on, and which customers are worth winning back?
James Wei	Head of Mer chandising	SKU-level profitability. Suspects some subcategories are high-volume / net-negative.	Which products should we pull from the catalogue and which should we push harder?

STEP 3 — BUSINESS QUESTIONS (10 Questions, 3 Tiers)

■ TIER 1 — DESCRIPTIVE

Q1 What is our actual P&L; by segment and category — including every loss-making order?

STAKEHOLDER	Linda Park (CFO)
DECISION	Identify which segment is subsidising the others and which category is the biggest margin leak before the board review.
METRICS	• Gross Revenue = SUM(sales) — all orders, no filter • Gross Profit = SUM(profit) — includes negative values; filtering them out is a material error • Gross Margin % = SUM(profit) / SUM(sales) — display negative if negative • Loss Order Count = COUNT(*) WHERE profit < 0 • Loss Order Rate = COUNT(profit < 0) / COUNT(*) per cell
TECHNIQUE	Multi-table JOIN: ecom_sales → product (category), ecom_sales → customer (segment via mode — see Q5 note). GROUP BY segment × category. Conditional aggregation for loss count.
OUTPUT	Pivot table (3 segments × 3 categories) colour-coded by margin %. Companion bar chart: total profit per segment, stacked by category. Red bars = negative profit.
■ TRAP	*Filtering out negative-profit orders to 'clean' the data is the most common mistake. Those rows ARE the story Linda Park needs to see. Never filter profit < 0 from a P&L; view.*

Q2 What fraction of our customers actually come back — and how does repeat behaviour differ by segment?

STAKEHOLDER	Marcus Tan (Head of Growth)
DECISION	Determine whether the acquisition funnel or the retention loop is the bigger lever for revenue growth.
METRICS	• One-Time Rate = COUNT(customer_id WHERE distinct_orders = 1) / COUNT(DISTINCT customer_id) • Repeat Rate = 1 – One-Time Rate • Avg Orders Per Customer = COUNT(DISTINCT order_id) / COUNT(DISTINCT customer_id) per segment • CRITICAL edge case: one order_id with 10 line items = 1 purchase, not 10. Must COUNT(DISTINCT order_id), never COUNT(row_id).
TECHNIQUE	CTE 1: COUNT(DISTINCT order_id) per customer_id. CTE 2: label single vs. repeat. CTE 3: aggregate by primary segment (use mode — see Q5). Window function not required but a CASE WHEN on the order count is.
OUTPUT	Funnel chart: 1 order → 2 orders → 3+ orders per segment. Summary table showing repeat rate and avg frequency side-by-side.
■ TRAP	*Using COUNT(row_id) instead of COUNT(DISTINCT order_id) inflates purchase frequency by the average items-per-order, making the business look 3-5× more sticky than it actually is.*

Q3 Which subcategories rank highest by revenue — and do they rank the same by profit?

STAKEHOLDER	James Wei (Head of Merchandising)
DECISION	Build the Q1 catalogue review list: which subcategories to promote, which to deprioritise, which to kill.
METRICS	• Revenue Rank = RANK() OVER (ORDER BY SUM(sales) DESC) per subcategory • Profit Rank = RANK() OVER (ORDER BY SUM(profit) DESC) per subcategory • Rank Delta = Revenue Rank – Profit Rank. Negative = more profitable than volume suggests. Positive = volume disguises poor margin. • Total Profit Destroyed = SUM(ABS(profit)) WHERE profit < 0 per subcategory
TECHNIQUE	CTE aggregate ecom_sales JOIN product → subcategory. Apply two independent RANK() window functions. Compute delta. Sort by Profit Rank, highlight where abs(delta) > 5.
OUTPUT	Scatter plot: x = Total Revenue, y = Total Profit, bubble size = Quantity Sold, colour = category. Four quadrants: Star / Cash Cow / Question Mark / Dog (BCG-style labelling).
■ TRAP	*Assuming high-revenue subcategories are the most important to defend. Technology subcategories are likely high-revenue / low or negative profit due to heavy discounting. Without the rank delta, merchandising doubles down on the worst offenders.*

■ TIER 2 — DIAGNOSTIC

Q4 Does discounting actually generate incremental volume, or does it only destroy margin?

■ NON-TRIVIAL — requires window functions, multi-step SQL, or statistical reasoning

STAKEHOLDER	Linda Park (CFO) — core question blocking the $400K campaign budget
DECISION	Approve, reject, or redesign the Q1 discount campaign. This is the single most important deliverable.
METRICS	• Discount Band = CASE WHEN discount = 0 THEN 'No Discount' WHEN discount <= 0.20 THEN '1-20%' WHEN discount <= 0.40 THEN '21-40%' WHEN discount <= 0.60 THEN '41-60%' ELSE '61%+' END • Avg Quantity per Line = AVG(quantity) per band × category • Avg Profit per Line = AVG(profit) — must remain signed; do not use ABS • Loss Rate = COUNT(*) WHERE profit < 0 / COUNT(*) per band × category • Profit Yield = SUM(profit) / SUM(sales) per band × category — the margin left after discounting
TECHNIQUE	CASE WHEN to build discount band. GROUP BY category × discount_band (not overall — Technology and Office Supplies have fundamentally different base margins; pooling them creates Simpson's Paradox). Conditional aggregation for loss rate. DO NOT use a single GROUP BY: category must be a control variable throughout.
OUTPUT	Multi-line chart per category: x = discount band, y-left = avg profit per line, y-right = avg quantity. Separate panel per category. Annotate the breakeven discount rate where avg profit crosses zero.
■ TRAP	*Concluding 'discount drives volume' from a pooled analysis. Technology has both higher base volume and heavier discounting — its volume signature will swamp the pooled result and manufacture a spurious positive correlation. Always stratify by category.*

Q5 Which segment produces the most profitable customers on a per-head basis — and what is the distribution of customer-level profit?

■ NON-TRIVIAL — requires window functions, multi-step SQL, or statistical reasoning

STAKEHOLDER	Linda Park (CFO) + Marcus Tan (Growth) — segment P&L; and acquisition targeting
DECISION	Where to allocate Q1 acquisition budget: acquire more Corporate accounts or more Consumer accounts?
METRICS	• Customer Lifetime Profit = SUM(profit) per customer_id across all orders — includes negative contributors • Median Customer Profit per segment (use PERCENTILE_CONT not AVG — skewed by a small set of large corporate accounts) • % Customers with Negative Lifetime Profit = COUNT(ltv < 0) / COUNT(*) per segment • SEGMENT ASSIGNMENT RULE: a customer_id may appear under multiple segments across orders. Assign the mode segment (most frequent) per customer. Document this assumption. Run a sensitivity check: does the P&L; ranking change if you use 'most recent segment' instead?
TECHNIQUE	CTE 1: COUNT(segment) per customer_id, use ROW_NUMBER() OVER (PARTITION BY customer_id ORDER BY seg_count DESC) to identify mode. CTE 2: SUM(profit) per customer_id. CTE 3: JOIN CTEs 1+2 with customer table. CTE 4: aggregate distributions per segment. Requires window functions, a subquery for mode logic, and careful join sequencing.
OUTPUT	Box plot: customer lifetime profit distribution per segment (show outliers). Second panel: heatmap — education_level × annual_income_band → avg customer profit (controls for observable selection effects).
■ TRAP	*Joining ecom_sales directly to customer and grouping by segment.sales without resolving the many-to-one customer-segment ambiguity. One customer with 10 Corporate orders and 1 Consumer order counts as 1 Corporate customer and 1 Consumer customer, inflating Corporate's apparent customer count by all switchers.*

Q6 Which markets are growing and which are saturating — and how does growth rate interact with profitability?

■ NON-TRIVIAL — requires window functions, multi-step SQL, or statistical reasoning

STAKEHOLDER	Marcus Tan (Head of Growth) — top-3 markets for expansion investment
DECISION	Allocate the Q1 expansion budget to three specific markets before the board review.
METRICS	• Annual Revenue per Market = SUM(sales) GROUP BY market, YEAR(order_date) • YoY Growth Rate = (revenue_year_N - revenue_year_N-1) / revenue_year_N-1 — use LAG(1) over market partition ordered by year. Note: single-year growth is noisy; compute for all available years and use the most recent full year. • Market Profit Margin = SUM(profit) / SUM(sales) per market per year • Market Size Weight: a 300% growth rate on $5K base is noise. Weight growth rate by absolute revenue when ranking markets. • New Customer Rate per Market = COUNT(DISTINCT customer_id with first_order in that market this year) / COUNT(DISTINCT customer_id) — requires subquery for first-order date
TECHNIQUE	CTE 1: SUM(sales), SUM(profit) per market per year. CTE 2: LAG(annual_revenue, 1) OVER (PARTITION BY market ORDER BY year). CTE 3: compute growth rate, margin, join to get absolute size. Final: 2×2 scatter. Requires LAG window function — not achievable with GROUP BY alone.
OUTPUT	2×2 scatter: x = Profit Margin %, y = YoY Growth Rate. Bubble size = absolute revenue. Quadrant labels: Invest (high growth, positive margin) / Harvest (low growth, high margin) / Fix (high growth, negative margin) / Exit (low growth, negative margin).
■ TRAP	*Ranking markets by growth rate alone. A market growing 200% YoY on a $10K base will outrank a market growing 15% on a $500K base. The second market represents 10× more absolute opportunity. Always plot growth against absolute size simultaneously.*

Q7 — Within Technology — the highest-revenue category — which subcategories destroy the most profit, and what is the structural discount driving it?

STAKEHOLDER	James Wei (Head of Merchandising)
DECISION	Design category-specific discount caps to include in the Q1 campaign policy.
METRICS	• Median Discount Rate per subcategory = PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY discount) — use median not mean; a few 80% discounts skew AVG dramatically • Total Profit Destroyed = SUM(ABS(profit)) WHERE profit < 0 AND category = 'Technology' • Profit Loss Rate = SUM(ABS(profit) WHERE profit < 0) / SUM(sales) per subcategory — the proportion of revenue that converts to destroyed profit • Discount Breakeven = the discount rate above which avg profit turns negative per subcategory (interpolate from the band analysis in Q4)
TECHNIQUE	Conditional aggregation with PERCENTILE_CONT. Filter category = 'Technology'. GROUP BY subcategory. Sort by profit loss rate DESC.
OUTPUT	Stacked horizontal bar chart: positive profit (blue) stacked with profit destruction (red). Sort by total profit destruction. Overlay median discount rate as a dot on a secondary axis.
■ TRAP	*Using AVG(discount). A subcategory might have 80% of orders at 15% discount and 20% of orders at 75% discount — the mean reads ~27%, masking that the tail is where all the destruction happens. Median gives the 'typical' order; a histogram would be even better.*

■ TIER 3 — PREDICTIVE / OPTIMIZATION

Q8 Which customers are at risk of churning in the next 90 days — and how much revenue are they collectively worth?

■ NON-TRIVIAL — requires window functions, multi-step SQL, or statistical reasoning

STAKEHOLDER	Marcus Tan (Head of Growth)
DECISION	Trigger a targeted win-back campaign before the customer goes dark. Prioritise by revenue at risk.
METRICS	• Inter-Purchase Interval = DATEDIFF(day, LAG(order_date) OVER (PARTITION BY customer_id ORDER BY order_date), order_date) — only for customers with >= 2 orders • Segment Avg Interval = AVG(inter_purchase_interval) per primary segment — computed only on customers with >= 3 orders to avoid noise • Days Since Last Order = DATEDIFF(day, MAX(order_date), [max_date_in_dataset]) • Churn Risk Tier: GREEN = days_since < 0.75× segment_avg; AMBER = 0.75-1.5× segment_avg; RED = > 1.5× segment_avg and < 3× (likely at-risk); GREY = > 3× (likely lost already) • Revenue At Risk = COUNT(RED customers per segment) × AVG(lifetime sales per segment) • Edge case: customers with 1 order have no interval — assign segment average as their expected interval. Flag them separately as 'unknown rhythm'.
TECHNIQUE	CTE 1: LAG window function to compute inter-purchase gaps per customer. CTE 2: AVG gap per segment on customers with >= 3 orders. CTE 3: MAX(order_date) per customer. CTE 4: join CTEs 2+3 → compute days_since / avg_interval ratio → assign tier. At minimum 4 CTEs with window functions. Cannot be done with a single GROUP BY.
OUTPUT	Table: customer_id, segment, last_order_date, days_since_last_order, churn_risk_tier, avg_order_value. Sort: RED first, then by avg_order_value DESC. Summary KPI: 'Total revenue at risk: $X across N customers.'
■ TRAP	*Using a flat 90-day threshold for all segments. Corporate customers commonly have 120-day purchase cycles (quarterly procurement). A flat 90-day rule would flag every Corporate account as at-risk when they are perfectly healthy. Thresholds must be segment-specific and derived from the actual purchase rhythm in the data.*

Q9 What does the RFM landscape look like — and which RFM segments are generating disproportionate profit vs. disproportionate revenue?

■ NON-TRIVIAL — requires window functions, multi-step SQL, or statistical reasoning

STAKEHOLDER	Marcus Tan (Growth) + Linda Park (CFO) — joint decision on Q1 retention spend allocation
DECISION	Assign retention budget across RFM segments: invest in Champions, reactivate At-Risk, or write off Lost customers?
METRICS	• R (Recency) = DATEDIFF(day, MAX(order_date), reference_date) per customer — lower is better • F (Frequency) = COUNT(DISTINCT order_id) per customer — higher is better • M (Monetary) = SUM(profit) per customer — USE PROFIT, NOT SALES. A customer who buys $10K/year at 70% discount is a net destroyer of value, not a Champion. • R_Score = NTILE(4) OVER (ORDER BY recency ASC) — score 1 = most recent (best) • F_Score = NTILE(4) OVER (ORDER BY frequency DESC) — score 1 = most frequent (best) • M_Score = NTILE(4) OVER (ORDER BY monetary DESC) — score 1 = highest profit (best) • RFM_Segment: Champions = R=1,F=1,M=1; Loyal = R<=2,F<=2; At Risk = R=3-4,F<=2; Lost = R=4,F=3-4 • CRITICAL: customers with negative lifetime profit must be included and will cluster in low M-score buckets — this is actionable information (stop spending on them).
TECHNIQUE	Three parallel CTEs computing R, F, M independently. Fourth CTE joins all three using customer_id. Fifth CTE applies NTILE(4) to each metric. Sixth CTE labels RFM segments using CASE WHEN on score combinations. Final query computes revenue, profit, and customer count per segment. Requires 5-6 CTEs and three window functions.
OUTPUT	Heatmap: 4×4 grid (R_Score × F_Score), cell colour = avg M (profit). Secondary table: RFM segment label, customer count, avg profit per customer, total revenue, % of total profit — sorted by total profit contribution. Highlight cells where revenue rank and profit rank diverge.
■ TRAP	*Using SUM(sales) for M. This is the most common RFM error in e-commerce datasets with heavy discounting. A customer who generates $50K in revenue at 65% average discount has negative lifetime profit — labelling them a Champion and spending retention budget on them wastes money. Always use profit as M when discount data is available.*

Q10 — For Corporate customers, which category combinations are bought together — and do multi-category buyers have meaningfully higher LTV?

■ NON-TRIVIAL — requires window functions, multi-step SQL, or statistical reasoning

STAKEHOLDER	James Wei (Merchandising) + Marcus Tan (Growth) — cross-sell email sequence design for Q1
DECISION	Build a category-level cross-sell recommendation for Corporate accounts: which category should we pitch first, and which should follow?
METRICS	• First Category = FIRST_VALUE(category) OVER (PARTITION BY customer_id ORDER BY order_date) — the entry category for each customer • Category Breadth = COUNT(DISTINCT category) per customer • Customer LTV = SUM(profit) per customer (signed — negative is negative) • Cross-Sell Pair Frequency = COUNT(DISTINCT customer_id) who purchased both category A and category B (self-join on customer_id with category filter) • LTV Lift = AVG(LTV for 2-category buyers) / AVG(LTV for 1-category buyers) — expressed as a multiplier (e.g., '2-category buyers have 2.4× the LTV of single-category buyers') • Edge case: only include Corporate-segment customers. Do not pool with Consumer — buying behaviour and category mix are structurally different.
TECHNIQUE	CTE 1: first category per customer (FIRST_VALUE window function). CTE 2: all categories per customer (string_agg or separate rows). CTE 3: self-join ecom_sales on customer_id to find co-occurring category pairs. CTE 4: LTV per customer. CTE 5: join LTV to category breadth and first-category. Aggregate: avg LTV by first_category × breadth. This is a multi-step SQL requiring a self-join plus two window functions.
OUTPUT	Two panels. Left: bar chart — avg LTV by first category × category breadth (1, 2, 3). Right: matrix table — entry category (rows) × second category (columns) → count of customers who made that transition. Highlight the highest-frequency cross-sell path.
■ TRAP	*Attributing higher LTV of multi-category buyers to the cross-sell itself rather than to underlying company size. A large corporate account buys across all three categories because they are large — not because they were cross-sold. Must control for annual_income as a proxy for company size. If income is not available at the required granularity, flag this as a confound that limits causal inference.*

STEP 4 — WHAT THIS DATASET CANNOT ANSWER

Business Question	Missing Data	What to Acquire
Does discounting increase conversion rate?	No browse/session/impression data	Web analytics: GA4, Amplitude, or Mixpanel event stream
What is CAC per segment?	No marketing spend data	Ad platform exports: Meta Ads, Google Ads, LinkedIn (campaign-level cos
What is the return/refund rate by product?	No return flag or credit note records	OMS (Order Management System) return data, joined on order_id
Does shipping speed affect repeat purchase rate?	No ship_date, only order_date	Logistics partner API or warehouse system
Are we price-competitive vs. the market?	No external pricing data	Price intelligence tool (e.g. Prisync, DataWeave) or manual scrape
What is customer satisfaction / NPS by segment?	No survey or review data	Qualtrics, Medallia, or star-rating data from platform
Is the discount effect causal or merely correlational?	No experimental design in data — all observational	A/B test: randomise discount rate at customer-level for 60 days

Assumptions to declare out loud in every deliverable:

1. discount is a rate (0.0–1.0). Verify with MIN/MAX before proceeding.
2. annual_income is in USD. No currency column in schema.
3. Canonical segment per customer = mode (most frequent) across order history. Sensitivity check: rerun with 'most recent segment' and report if ranking changes.
4. Churn thresholds are segment-specific (derived from inter-purchase interval distribution), not a flat 90 days.
5. profit in the dataset is net of COGS and logistics fees — not gross profit. If this assumption is wrong, all margin calculations are materially overstated.
6. The analysis treats the dataset as a complete population, not a sample. No confidence intervals are applied to totals.

STEP 5 — PROJECT STRUCTURE & DELIVERABLE STACK

1	Data Audit Memo	0.5 days	Confirm date range. Verify product_code uniqueness. Count orphaned FKs. Validate discount fo
2	KPI Definition Layer (kpi_definitions.sql)	0.5 days	SQL views or CTEs that define exactly 8 core metrics: Gross Revenue, Gross Profit, Gross Marg
3	Analysis Notebooks (4 files)	3–4 days	01_baseline.sql (Q1–Q3) · 02_discount_analysis.sql (Q4, Q7) · 03_customer_analytics.sql (Q5,
4	Dashboard (5 views)	1.5 days	P&L Overview (CFO) · Discount Impact Matrix (CFO + Merchandising) · Market Heatmap (Grow
5	One-Page Recommendation Memo	1 day	Format: Situation → Finding → Recommendation → Expected Impact (quantified). Target output

STEP 6 — INTERVIEW READINESS

Interview Question 1	**Walk me through your RFM model. Why did you use profit instead of revenue for the M dimension?**
WEAK ANSWER	*I used NTILE(4) for R, F, and M then summed the scores to get a composite. Revenue is the standard M metric.*
STRONG ANSWER	I used profit as M because this dataset has discounts ranging up to 80% — a customer generating $50K revenue at 65% average discount is a net destroyer of value, not a Champion. Using revenue as M would cause me to label the worst-margin customers as Champions and recommend spending retention budget on them. Direction also matters: R scores are inverted (lower recency = better), so I reversed the NTILE ordering for that dimension. I documented both choices explicitly so the model can be re-run with revenue M as a sensitivity check.

Interview Question 2	**You assigned each customer a 'primary segment' using mode. How confident are you that this is the right approach?**
WEAK ANSWER	*It seemed like the most logical approach since segment is at the order level.*
STRONG ANSWER	It's the most defensible approach given the data, but it has real limitations I declared in the memo. A customer with 10 Corporate orders and 3 Consumer orders gets labelled Corporate — which is likely correct. But a customer with 5 Corporate and 5 Consumer orders is genuinely ambiguous: they might be an individual who also orders for work. I ran a sensitivity check using 'most recent segment' instead of mode. If the P&L; ranking by segment changes materially between the two methods, that itself is a finding worth surfacing — it means segment classification is unstable and the business should fix the data collection.

Interview Question 3	**[METHODOLOGY ATTACK] Your discount analysis shows lower profit at higher discount bands. But you can't actually conclude that discounting causes margin destruction from this data. What are the three most plausible alternative explanations?**
WEAK ANSWER	*You're right, it's correlational. I'd need an A/B test to confirm.*
STRONG ANSWER	Three specific confounders. First, selection into discounting: marketing may specifically apply heavy discounts to clearance items or products with intrinsically low base margins — the discount is a response to low margin, not the cause of it. Second, category mix: Technology both has higher base volume and receives heavier discounts; if Tech has structurally lower margins than Office Supplies, the discount-margin correlation in pooled data is partially a category-composition effect. Third, customer segment sorting: Corporate customers may negotiate bulk discounts while also having higher order values — the segment drives both the discount and the volume. I partially address confounders one and two by stratifying every analysis by category. Confounder three requires the per-customer LTV analysis in Q5 to disentangle. True causal identification would require a randomised discount experiment — I recommend this as the Q2 follow-on.

Interview Question 4	**Your churn analysis uses segment-specific inter-purchase intervals as thresholds. But you computed those averages from the same population you're scoring. Isn't that circular?**
WEAK ANSWER	*That's a fair point but there's no other data to use.*
STRONG ANSWER	It is partially circular, and I flagged it in the methodology. The best fix would be to train the threshold on a historical cohort where we know the eventual outcome — i.e., use the 2022 cohort to find the inter-purchase interval that best separates 'eventually returned' from 'truly churned,' then apply that threshold to the current population. Without ground-truth churn labels, I use a partial fix: I compute the average interval only on customers with 3 or more orders. This excludes customers who may have bought twice by coincidence, using only customers who have demonstrated a genuine purchase rhythm. It reduces but does not eliminate the circularity, and I say that in the memo. I also flag that the threshold should be recalibrated annually as the customer base grows.

Interview Question 5	**If you had to present only one number to the board to justify your recommendation on the discount campaign, what would it be and why?**
WEAK ANSWER	*Overall gross margin or total profit.*
STRONG ANSWER	The loss order rate at discount bands above 40%, broken out by category. Specifically: 'X% of Technology orders with discount > 40% are loss-making, representing $Y in destroyed profit per year on $Z in revenue.' That number directly answers Linda Park's question — is discounting net-positive? — and it's specific enough to drive a decision: either cap discounts at 30-35% in Technology, or discontinue Technology discounting entirely. Any other number — total revenue, overall margin — is too aggregated to create urgency. One metric, one category, one discount threshold: that's the slide the CFO can act on.